

TOPIC 4.2

What do we know about the universe based on what we can see only with our eyes?

Key Concepts

- We see a variety of objects that appear in the sky in mostly predictable ways.
- The appearance and motions of the objects we see can be described using a model called the celestial sphere.

Curricular Competencies

- Experience and interpret the local environment
- Apply First Peoples perspectives and knowledge, other ways of knowing, and local knowledge as information sources
- Evaluate the validity and limitations of a model or analogy
- Construct, analyze, and interpret models and diagrams

The sky is a part of our local environment wherever we live. For many, including First Peoples, knowledge of the observable universe is part of an interconnected world view that includes social, cultural, and spiritual aspects. This photo of a well-known medicine wheel in Wyoming is one example of how First Peoples connect with their world view. First Nations of the North American plains built hundreds of medicine wheels. Most are in Canada. The largest and oldest, in Alberta, dates back at least 5000 years. Many align with the Moon, Sun, and stars, and predict events such as solstices and the rising of sky objects at certain times of the year. Because people observe the sky from their local place, there is a great diversity of perspectives about the universe. All are respected.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

- 1. Applying First Peoples Perspectives** First Peoples knowledge of space comes from observations specific to the places where they live. For example, medicine wheels developed in flat open prairies, but not in mountainous British Columbia. Find examples of connections between Indigenous knowledge about the universe and the local environment. Look for traditional stories, or information about how reading the skies helped people to survive. 
- 2. Questioning** Compare the title of this Topic with that of Topic 4.3. How does technology affect the ways we think about and interpret what we can observe?
- 3. Identifying Preconceptions** Some human societies, of the past and in modern times, construct their calendars based on patterns related to the Sun. Other societies have used and continue to use the Moon. What are the advantages and disadvantages of using either or both the Sun and the Moon to define the cycle of the year? Which societies and peoples use either or both today?

Key Terms

There are three key terms that are highlighted in bold type in this Topic:

- constellation
- celestial sphere
- ecliptic

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

We see a variety of objects that appear in the sky in mostly predictable ways.

The things that you can see in the night sky, tonight, are the same as those that the first humans saw when they themselves looked up. Perhaps people with finely tuned eyesight can make out a few more details or even notice a dim star that you can't. However, until the telescope was turned to the sky in the early 1600s, the sky of 10 000 years ago and the sky of today mostly reveal the same objects and events. What are these?

The activity that fills the rest of this Concept tours and provides questions about some of the sky objects that are visible to the unaided eye. You will also be invited to add your own objects and questions.

Activity

A Tour of Visible Sky Objects



Use the questions below and the questions that appear with each photo to share, collect, and evaluate your ideas about visible sky objects.

Questions To Guide and Inspire Ideas

- How might place affect what you see in the sky? For example, how might observations from Prince Rupert compare with Yellowknife, Grand Forks, Tucson, or Mexico City? What if you lived in a valley, on the coast, or at a high elevation?
- What questions would you ask about these objects if you lived 10 000 years ago?
- What stories, past and present, do we tell about these objects and events, and how do they compare?

The Sun

The most obvious daytime object—the “ruler” of the daytime sky—is the Sun. **How does it appear to move in the sky? What path does it follow? Is it always visible each day? Why is the Sun equated with leadership and power in so many cultures?**





The Stars

On a clear, dark night, you can see about 5000 stars.

How do stars compare with one another in terms of location, colour, and brightness? How do they appear to move? Is there anything significant about the path they follow? Is there something special about the star in the north that never seems to move at all? Why do stars disappear when the Sun comes out? What patterns in appearance and movement are there, and how could they be understood or used?



The Moon

The only moon that is visible to the unaided eye is our own. How does the Moon change from night to night? How does its movement compare with the Sun? What patterns and movements can we observe? How can we use these observations?



Five Planets

The only planets visible to the unaided eye are Mercury, Venus, Mars, Jupiter, and Saturn. Are they always visible? How do they move, and how does that compare with stars and other sky objects?

Over To You

Other objects and events that are visible to the unaided eye include comets, meteorites (shooting stars), “guest stars” (novas and supernovas), and the Milky Way. Are there others? Find images, and write one or two sentences to describe them. Then record questions (and answers) like those in the rest of this activity.



Before you leave this page . . .

1. How might people of the past have thought about the “universe” based on the objects and events they could see in their skies?



Figure 4.3 The pose of the human figure on this ancient ivory is similar to the pose of the figure that later cultures saw in the stars of Orion.

Evidence of People's Observations of Sky Objects Long Ago

How do we know that people of the past observed and recorded the sky objects you are now familiar with from the previous page? We know because they left evidence of what they saw and studied. You saw one example of this—a medicine wheel—at the start of this Topic. **Figure 4.3** shows another example. This tiny piece of ivory from 32 000 years ago records a glimpse of what its maker saw in the sky. The mammoth tusk, found in Germany, is thought to show a pattern of stars in the night sky—possibly the constellation we call Orion.

Another example was left by ancient cousins of ours, the Cro-Magnon people, who lived in Europe more than 15 000 years ago. See **Figure 4.4**. They used their observations of the changing night sky to draw the first-known lunar calendar on cave walls near Lascaux, France. The 29 dots represent the 29 days of the Moon's cycle of phases. (It takes 29.5 days for the Moon to complete a full cycle from one new Moon to the next.)



Figure 4.4 This stunning painting is one of about 2000 made deep in caves in southern France 15 000 to 17 000 years ago.

One of the most famous monuments in the world is Stonehenge on England's Salisbury Plains (**Figure 4.5**). Although much of its purpose continues to be studied, the structure suggests one function was as an astronomical observatory. Built starting about 5000 years ago, Stonehenge is a circular arrangement of giant stones and boulders used to mark the movement of the Sun, the Moon, and their eclipses. The entrance aligns with the rising Sun on the summer solstice (longest day of the year).



Figure 4.5 Among its functions, some understood and some not, Stonehenge is a type of astronomical observatory, with key stones in alignment with the rising Sun and other sky objects on specific days and nights of the year.

In nearby Ireland, and dating from roughly the same time, is the tomb monument of Newgrange. As shown in **Figure 4.6**, the rising Sun on the winter solstice aligns with a passage and chamber built within this 80 m-diameter structure.



Figure 4.6 A UNESCO World Heritage Site, Newgrange is located about an hour car-drive north of Dublin in a place called Brú na Bóinne. People were farming and building great monuments here at least half a millennium before the great pyramids were being built in Egypt.

Records of sky observations were not limited to the stars, Sun, and Moon. The Maya of Central America are renowned for their meticulous observations of planets. For example, the Dresden Codex, shown in **Figure 4.7**, includes detailed mathematical calculations for the planet Venus.



Figure 4.7 The Maya wrote in “fold-out” books called codices (singular codex). Of the thousands that existed before European contact, only four remain. The rest were burned. **Questioning:** What things—both material and otherwise—were lost by destroying the Maya codices?



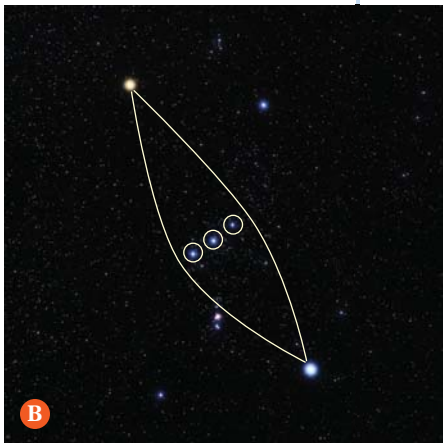
Before you leave this page . . .

1. How do we know that people of the past observed the movements and patterns of sky objects?
2. How do you think people knew exactly where the Sun and other sky objects would rise on a particular day of the year?

CONCEPT 2

The appearance and motions of the objects we see can be described using a model called the celestial sphere.

Figure 4.8 The stars that make up the constellation officially named Orion are shown in **A**. In **B**, the same stars are shown in the shape of a canoe.



Imagine you are a hunter thousands of years ago. You know the patterns of the night sky the way you know the land on which you live and the habits of the plants and animals that sustain you. On this day, you have followed the trail of animals to a place that is unfamiliar to you. As night approaches, you locate a safe spot, unfurl your bed roll, and lie back to stare up at the sky. Though the land on which you rest is new to you, the star-scape that twinkles down at you is much like what you know. The stars will guide you when it is time to return home. For now, their familiar patterns provide comfort. Beneath this sky you know, your sense of place, of home, is reinforced.

Many peoples, both past and present, connect objects and events on Earth with objects and events in the sky. For example, many First Peoples noted that the appearance of certain patterns of stars and certain “faces” of the Moon marked the changing seasons. This helped them know when to plant and harvest crops, perform ceremonies, and carry out other aspects of life.

Stars were also used for navigation. People living in places surrounded by water had no land to use as a reference point, so stars were used as familiar “landmarks.” Polynesians, for example, navigated the vast expanses between Pacific islands because of their understanding of stars, as well as of tides and currents.

Constellations

Over millennia, many cultures have observed that some stars in the night sky appear to form shapes. They named these shapes after heroes, animals, tricksters, and objects from their land, life, and stories. Today we call them **constellations**.

People used their imaginations to link up the stars like a huge connect-the-dots puzzle. Different cultures linked up the stars in different ways. For instance, ancient Greek skywatchers saw the stars in **Figure 4.8A** as making up the shape of Orion, a mighty hunter from their stories; the three stars at his waist are his belt. Some First Peoples saw in that same constellation and three-star grouping the shape of a canoe (**Figure 4.8B**). In southern Mexico, the Maya of a great city-state, Palenque, saw a turtle with the three stars forming part of its shell pattern. In the north the Inuit saw, instead, a bear chased by four hunters. The topmost star is the bear escaping by climbing high in the sky. Three hunters are in pursuit. The fourth hunter returned to Earth to retrieve a dropped mitten and remained to tell the story of the constellation.



constellation group of stars that form a pattern in the sky

Activity

Construct a Model Constellation

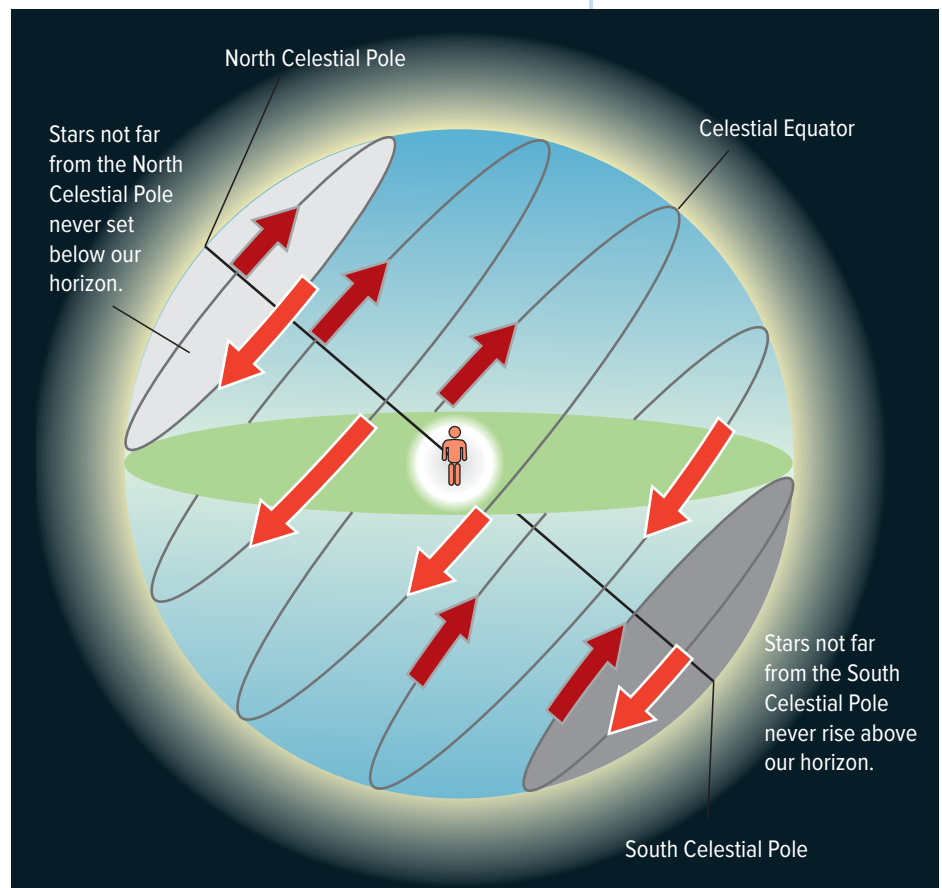
In the sky, the stars of a constellation look like they are all close to each other and at exactly the same distance from Earth. In fact, stars are separated from us and from each other by very large distances. Design a model to show how this is possible. Your model could be 3D, or you could render it on paper or digitally.

The International Astronomical Union lists 88 constellations. Orion is one of them. Smaller groups of stars that form shapes within the constellations are called *asterisms*, from the Greek word for star: *aster*. An example is one of the most famous and visible star patterns in the northern sky, the Big Dipper. The two end stars in its bowl are called “pointer stars”, because they point toward the North Star, Polaris (Figure 4.9). Long before the invention of the compass, people used the North Star to tell direction.

Here in Canada, we cannot see all 88 constellations. There are about 20 that never rise above our horizon. These 20 can be seen only by people who live (or visit) south of the equator. Use Figure 4.9 to figure out why.

When European explorers of the 15th century first sailed around the world and travelled to places like Australia and South America, they saw these stars for the first time. The explorers imagined new constellations out of these stars, and they chose objects that were important to their lives, such as a ship, a clock, a microscope, and a compass.

Figure 4.9 Imagine standing on flat land at night. As Earth turns, the stars appear to be on the inside surface of a sphere that moves slowly around Polaris, the North Star. You see stars in the sky above you. **Inferring:** Can you see the stars in the grey area? Why or why not?



The Celestial Sphere

celestial sphere an imaginary, rotating sphere on which lie all objects of the night sky

From Earth, we observe the sky as it appears, not as it is. For millennia, people believed the sky to be a solid sphere with celestial objects in fixed positions. This imaginary sphere onto which all celestial objects are projected is called the **celestial sphere**. Today, we know that it is Earth that rotates, and not the sky. The celestial sphere extends around Earth. However, an observer on Earth can only see half of the sphere, the same way you can only see what is in front of you, and not behind (**Figure 4.10**).

The celestial sphere is divided into northern and southern hemispheres by the celestial equator, which is an imaginary extension of Earth's equator projected into space. Astronomers mark the position of stars and other objects in the sky of each hemisphere using a grid system similar to latitude and longitude used on Earth. The North and South Celestial Poles are the imaginary points where Earth's axis of rotation extends out onto the celestial sphere above the North and South Poles on Earth. For observers in the northern hemisphere, the North Celestial Pole points very close to Polaris, the North Star. Even today, celestial navigation is widely used by sailors (**Figure 4.11**).

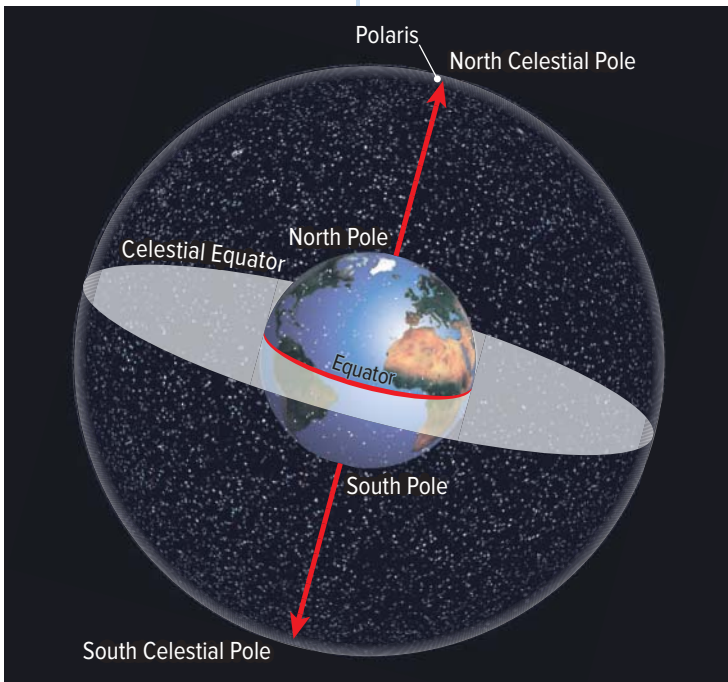


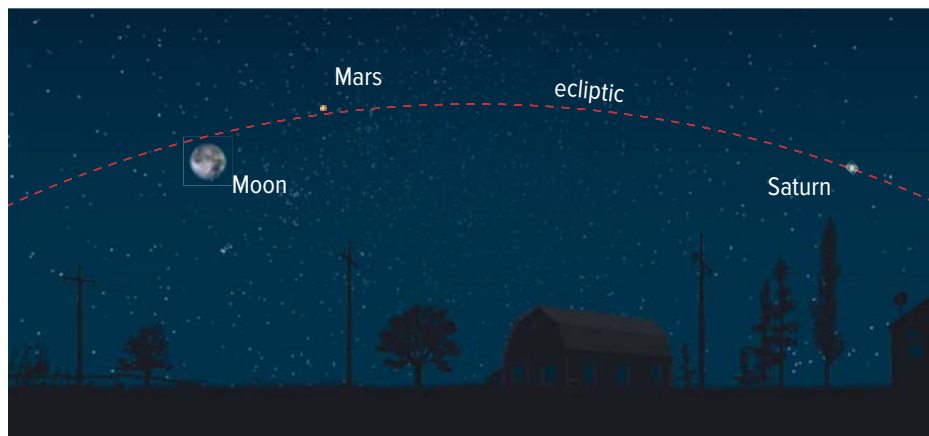
Figure 4.10 The celestial sphere **Analyzing:** How does this diagram compare with **Figure 4.9**? How does examining each help you understand the concept of the celestial sphere?

Figure 4.11 Modern technologies are widely used to navigate ships at sea. But batteries can run out, and GPS can malfunction. As a result, many navies and coast guards around the world have resumed learning the ancient art of celestial navigation, using an angle-measuring device called a sextant.



The Ecliptic

Over one year, the Sun seems to move against the constellations of the celestial sphere. The Sun's path across the celestial sphere is called the **ecliptic**, and is caused by Earth's orbit around the Sun. The ecliptic is best observed looking up at the sky toward the southern horizon (**Figure 4.12**).



ecliptic path the Sun and some other sky objects appear to take across the celestial sphere

Figure 4.12 The ecliptic is an important reference pathway that helps us locate the positions of the Sun, Moon, planets, and constellations. (The ecliptic gets its name from the similar-looking word, eclipse. Eclipses of the Sun and Moon occur along or near the ecliptic.)

Changing Views of the Night Sky

You know that Earth's rotation causes celestial objects to appear to move across the sky from east to west. The distance of celestial objects can also play a role in how they are observed from Earth. For example, because the planets are much closer to Earth than the stars, their path along the ecliptic over time appears to change with respect to the constellations. A planet can be distinguished from a star in the night sky by observing its motion over weeks or months.

In addition, our view of the night sky changes with each passing season due to Earth's revolution around the Sun. For example, if we observe the northern hemisphere's night sky in winter we see stars that are opposite the Sun. The stars that are in the same direction as the Sun will only be in the northern hemisphere's sky during the day and will not be visible (**Figure 4.13**). Therefore, the constellations we see in the night sky are not the same at all times of the year.

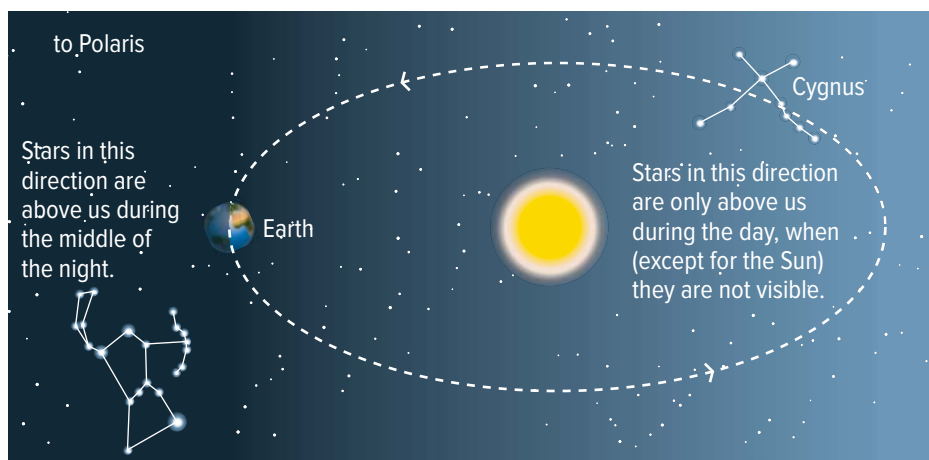
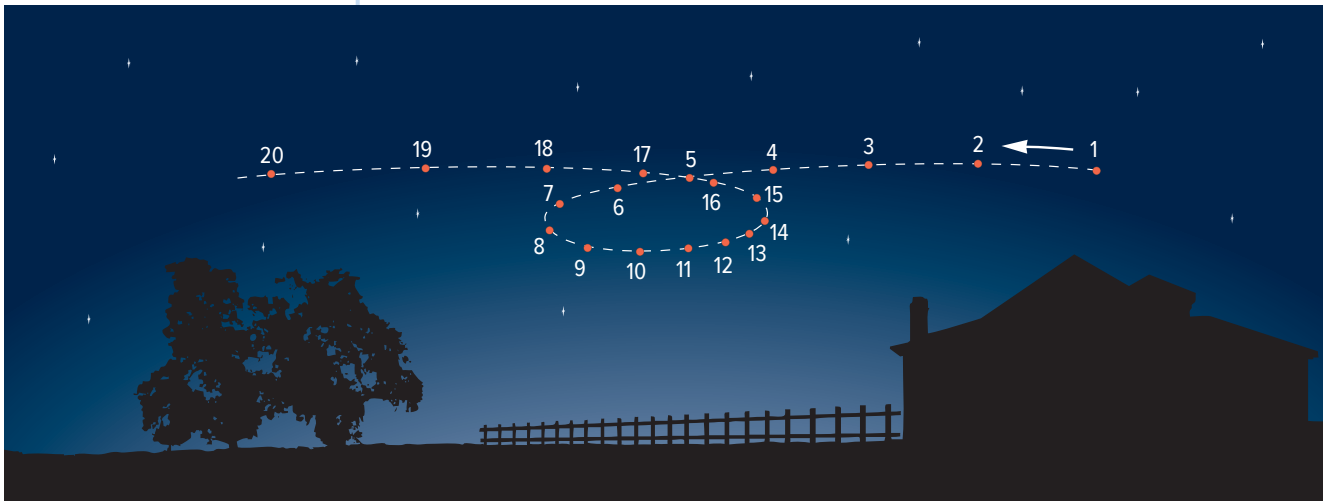


Figure 4.13 Orion is visible in the northern winter sky because it is opposite the Sun during that time. However, Cygnus is in the northern winter sky during the day, so it cannot be seen.

Retrograde Motion

Figure 4.14 The position of Mars compared to the background stars during a period of retrograde motion. Each dot represents the planet's new position every 10 days over the period.

Ancient astronomers observed that certain “stars” seemed to wander among the background of fixed stars. (These “stars” are really planets, from the Greek *planetes*, meaning wanderer.) As days pass, these objects appear to slow, stop, reverse direction, and loop back across the sky. This *retrograde motion* occurs because Earth orbits the Sun faster than the outer planets do. For example, as Earth passes Mars in its orbit, Mars seems to stop and then move backward. Earth continues past Mars, and forward motion appears again in the sky (**Figure 4.14**).



Activity

Demonstrate Retrograde Motion

1. Find a partner. One of you will represent Earth, and one will represent Mars.
2. Stand beside your partner. Mars walks slowly forward from the starting point. Earth stays at the starting point, pointing at Mars as Mars moves forward, as shown in the photo.



3. Earth leaves the starting point while still pointing at Mars, moving quickly to overtake the slower-moving Mars.

4. As Earth overtakes Mars, Earth begins to point backwards, as shown in the photo. This illusion of backwards motion is retrograde motion.



5. Switch roles, and repeat steps 2 to 4.
6. a) Why is retrograde motion visible only with planets farther from the Sun than Earth?
b) While in retrograde motion, in which direction does Mars appear to move with respect to the background stars?

Navigating the Night Sky

Finding your way around the night sky can seem confusing at first. The first step is to learn how to describe the location of celestial objects that you can see with the unaided eye. In geography, latitude and longitude are used to pinpoint a place or object on Earth. For navigating the sky, one method you can use involves celestial coordinates called azimuth and altitude to describe the position of a celestial object relative to an observer on the ground for a particular time and place.

Azimuth and Altitude

Azimuth is the distance measured from north along the horizon to a point directly below the celestial object. North has an azimuth of 0° , east has an azimuth of 90° , south has an azimuth of 180° , and west has an azimuth of 270° . Altitude is the angular height of a celestial object, measured from the horizon. **Figure 4.15** shows how the degrees of azimuth and altitude are measured.

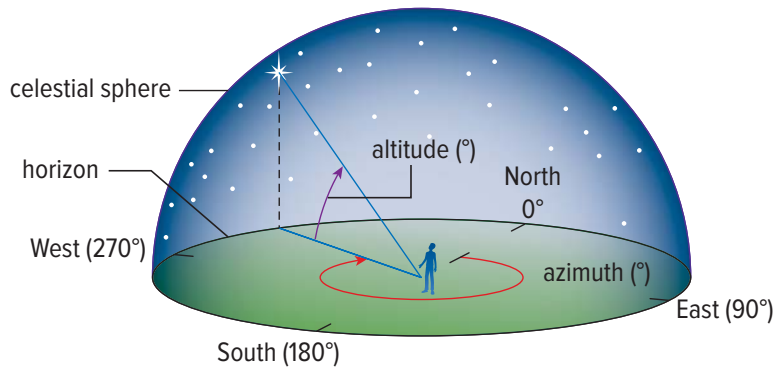


Figure 4.15 You measure azimuth and altitude with reference to the horizon.

You can use your hands to measure angles for determining the positions of celestial objects. For instance, when you hold your hand out at arm's length, the width of your index finger is approximately equal to 1° (**Figure 4.16A**). The width of a fist at arm's length is approximately equal to 10° (**Figure 4.16B**). If you extend the fingers of your hand, the width from the tip of your pinky finger to the tip of your thumb is approximately equal to 20° (**Figure 4.16C**). In this way, you can estimate the altitude and azimuth coordinates of any celestial object in the sky. Data obtained in this manner are relative to your position on Earth.

Figure 4.16 You can use your hand to estimate angles between objects in the sky.



Activity

Estimating Altitude and Azimuth

This activity is written using the Moon. However, your teacher may choose a different object depending on time of year and if the activity is being done during the day or evening. You will need a compass and notebook or digital recording device. There must be a trusted adult present as a safety precaution.

1. As your teacher instructs, go to a safe open area such as a park or parking lot.
2. Use a compass (or Polaris) to find north.
3. Start by facing north with your fists outstretched at eye level.
4. Keeping your arms straight out in front of you at eye level, use your fists as measuring tools while you slowly turn in a clockwise direction toward the east. Continue measuring until you reach the spot on the horizon directly below the Moon. Remember that each fist represents 10° .
7. With your arms still outstretched at eye level, begin counting degrees with your fist in an upward direction until the top of your fist reaches the Moon. Again, use your fingers to measure smaller increments as you get closer to the Moon. When your fist reaches the Moon, you have determined the altitude of the Moon from your position on Earth.



5. You can also use your fingers to measure smaller increments as you get closer to the Moon. Each finger width is 1° .
6. When you reach the spot directly below the Moon, stop counting. The value you have counted, in degrees, is the azimuth of the Moon from your location on Earth.
8. Record your results.
9. Compare your results with those of your classmates. Why do you think they are not exactly the same? Does that make this method invalid? Give reasons to justify your answer.
10. Would the altitude and azimuth of the Moon (or any other celestial object) be the same for people in different parts of the country? Why or why not?



Before you leave this page . . .

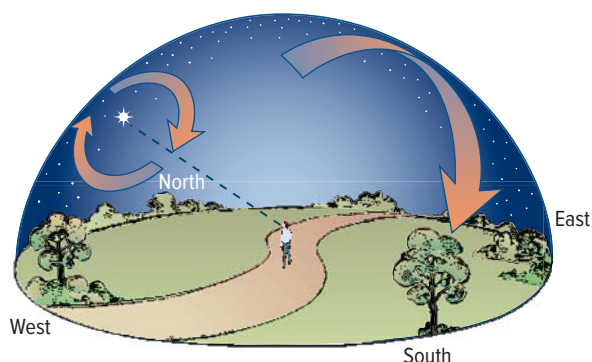
1. Using a globe as a guide, describe where you might find the celestial sphere, celestial equator, and celestial poles.
2. How can you distinguish a planet from a star in the night sky?
3. Use a sketch to explain the meaning of azimuth and altitude.
4. Describe how you can use your hands to determine the position of sky objects.

Check Your Understanding of Topic 4.2

QP Questioning and Predicting **PC** Planning and Conducting **PA** Processing and Analyzing **E** Evaluating
AI Applying and Innovating **C** Communicating

Understanding Key Ideas

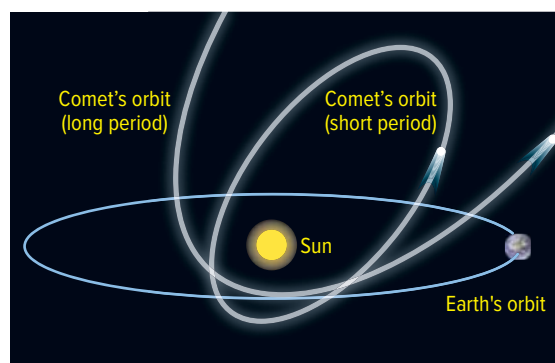
- Describe two ways that people have used motions of sky objects in relation to their daily lives. **PA**
- How are constellations and asterisms similar and different? **PA**
- Use the diagram to help you explain
 - the direction that stars in the night sky appear to move
 - why the star in the north has special significance to skywatchers who live north of the equator **QP PA**



- Everyone living in the northern hemisphere generally sees the same pattern of stars. If this is the case, why have the constellations and asterisms been interpreted in so many different ways? Give reasons for your views. **E AI C**
- The International Astronomical Union lists 88 officially recognized constellations. Why can you not see all of them in the Canadian sky? **PA**
- Some constellations represent signs of the zodiac and are associated with specific seasons. Why do we see different constellations at different times of the year? **QP PA**

Connecting Ideas

- What is the name of the path across the sky that the Sun seems to follow?
- The planets follow the same path through the sky that the Sun appears to follow. Unlike these objects, when comets are visible to us, they can appear anywhere in the sky. Use the diagram below to infer why. **AI**



- Imagine that you are lost in a desert at night. The desert is in the northern hemisphere. Explain how you could determine the four directions. **AI C**

Making New Connections

- When you observe the sky at night, you can see hundreds or even thousands of tiny specks of light. Some are brighter than others, and some appear larger than others. Thousands of years ago, ancient astronomers noticed that some of the objects were different than others. These were planets. Describe at least two observations that you could make without telescopes that would suggest planets are different from stars. **AI C**
- Would you search for a planet in the constellation Orion or Cygnus? Why or why not? **AI**

Ursa Major: How many stories over time and space?

What's the Issue?

The name Ursa Major is Latin for “great bear,” and the Romans inherited many of their tales about this constellation from the Greeks. However, stories from other ancient cultures saw different objects, animals, and beings in these stars. For example, Hindu astronomers saw the seven stars of the Big Dipper as seven great sages, ancient Egyptians saw a crocodile, and Babylonians saw a wagon. No doubt, many stories about this constellation have been lost to history. Perhaps they were never recorded orally or in writing. Perhaps they were never shared with anyone at all. We can, however, explore the lore and lessons that have been shared with us from various peoples of various times and places.

For example, in one First Peoples story, the stars forming the handle of the Big Dipper are three hunters chasing a bear. Because the stars are low in the sky during autumn, it is said that the hunters had injured the bear. Blood from the bear is turning the leaves red. Nova Scotia Mi'kmaq and St. Lawrence River Iroquois tell stories in which one of the handle stars is a hunter carrying a bow and arrow to kill the bear; another is a hunter carrying a pot to cook the bear in; and the third handle star is a hunter carrying firewood to use to heat the pot.

One of the ancient Greek stories told that Zeus, king of the Greek pantheon, changed a woman named Callisto into the great bear to save her from queen Hera. As a bear, Callisto was safe from Hera but was in danger of being killed by hunters. Callisto's son Arcas tried to kill her, not knowing the bear was his mother. Zeus changed Arcas into a bear too, to prevent him from killing Callisto. The pattern of stars representing Arcas is known as Ursa Minor, or Little Bear.

In many African cultures, the stars that make up the Big Dipper represent a drinking gourd. Africans who were enslaved in North America sang songs about the gourd and may have used the constellation to travel north to escape slavery.

An Arabic story interprets the stars in the Big Dipper's bowl as a coffin and the three stars in the handle as mourners following the coffin.

In China, the stars of the northern sky, including those of the Big Dipper, were part of a vast celestial empire that mirrored kingdoms and dynasties on Earth. Just as the terrestrial empire revolves around the emperor, the hub of the celestial empire was Polaris, around which all the northern stars circled.

From India, the seven stars of the Big Dipper were the Saptarishis, the seven semi-immortal sages whose stories appear notably in the longest-known epic in Earth's history, the *Mahabharata*.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. The names of the stars that make up the handle of the Big Dipper are Arabic: Alioth, Mizar, and Alkaid. The official names of many of the stars come from astronomers of the Islamic world. What role have they played in the history of astronomy, and what contributions have they made to our knowledge and understanding of the night sky?
2. The seven stars that make up the Pleiades (PLEE-ah-deez) are also known as the “seven sisters.” This grouping is not a constellation, but it is a powerfully evocative cluster of stars for many peoples through human history. What can you discover about the lore and lessons of the Pleiades?
3. Imagine that you have been asked to contribute to a new program for BC's Knowledge Network called *Star Wisdom Planetarium*. Find or develop a story (old or new) about Ursa Major or a constellation of your choice. Your story can be in the form of a narrative, graphic novel, song, mural, or other creative work.

Skills and Strategies

- Processing and Analyzing
- Evaluating
- Communicating
- Applying and Innovating

What You Need

- star map (from your teacher)
- ruler
- “Using Star Maps” in Appendix A

Using a Star Map

In this Investigation, you will learn how to use a star map. Star maps show the constellations and bright stars that are visible at a certain time of the year. The star map you will use can be used throughout the year and is specific to the northern hemisphere.

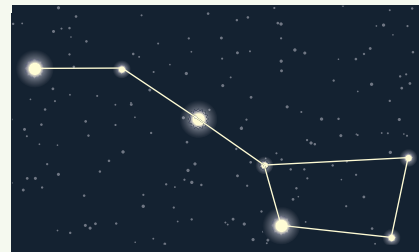
Question

How can you use the Big Dipper to locate other constellations?

Procedure

Part 1: Circumpolar Constellations

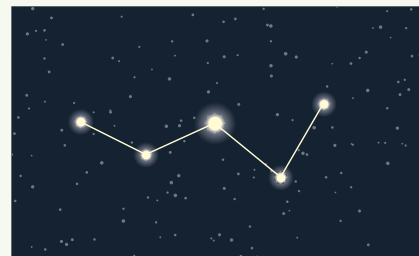
1. Look at the Big Dipper diagram below. Using a ruler, draw lines on your star map to join the stars of the Big Dipper.



2. Draw in the lines for the Little Dipper. Label the star at the tip of the handle “Polaris.” Polaris is also called the North Star, because you need to face the geographic North Pole to see it.

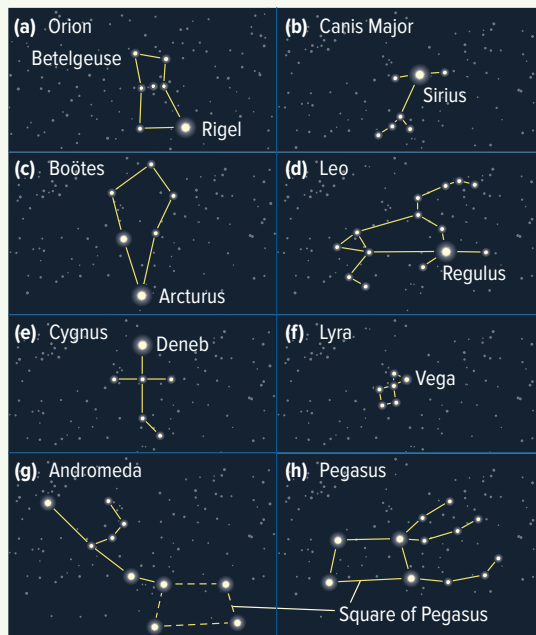


3. Locate the two stars of the Big Dipper farthest from the handle. These are called pointer stars, because you can use them to point to other constellations. Use them to draw a dashed line to Polaris.
4. Continue your dashed line until you reach Cassiopeia (Kah-see-oh-PEE-ah), which looks like a stretched-out W. Join its stars.



5. Label all three constellations on your star map.

6. Below are some other constellations visible during different seasons. Find the three stars in Orion that make up its belt. Draw in this constellation, which can be seen easily in December. Also label the stars Rigel and Betelgeuse (BEE-tuhl-juice).



7. Draw a dashed line to show how you would use Orion's belt to locate Sirius, the brightest star in the night sky. Draw Canis Major, which contains Sirius.
8. Locate and draw the constellation Boötes (boo-OH-teez) on your star map. To find it, first look for the star Arcturus, which is part of this constellation. Draw a dashed line to show how you would use the handle of the Big Dipper to find Arcturus.
9. Draw the constellation Leo on your star map. Leo contains the star Regulus. Draw dashed lines to show how you would use two stars of the Big Dipper as pointer stars to Regulus.
10. The three bright stars Deneb, Vega, and Altair form the Summer Triangle. Join them with dashed lines. Deneb and Vega are in the constellations Cygnus and Lyra. Draw these constellations on your star map.

11. Draw the lines for the remaining constellations on your star map. For each, use a dashed line from another constellation. When you have finished, share and compare your maps with the class.

Part 2: Locating the Zodiac Constellations

12. Notice there are still many unlabelled constellations on your star map. These are the zodiac constellations. They are visible during specific months, and the planets can be seen passing through them. You have already labelled one of the zodiac constellations, Leo. There are 11 more for you to find. The months in which they are best viewed are shown in the margin of the star map. Starting with Leo, label the rest of the zodiac constellations as follows: Virgo, Libra, Scorpius, Sagittarius, Capricornus, Aquarius, Pisces, Aries, Taurus, Gemini, and Cancer.

Conclude and Communicate

- Which constellations are visible from Canada in all seasons?
- Using your star map, identify two constellations that you can see from Canada during each of the four seasons.
- Describe how you used pointer stars in the Big Dipper to find three other constellations.

Apply and Innovate

- Look at your star map again. Can you find any other uses for pointer stars? Draw these uses with dashed lines, and describe them.
- To locate the constellations at night, you stand facing north and hold the star map with the current month at the top. Why do you hold the star map this way?

Skills and Strategies

- Questioning and Predicting
- Processing and Analyzing
- Evaluating
- Communicating

What You Need

- glue stick
- scissors
- copy of a star map and frame
- two file folders
- split pin

Modelling Motion in the Night Sky

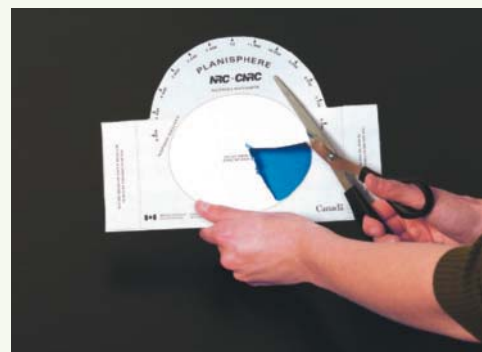
The motion of the stars in the night sky is caused by Earth's rotation around its axis and its revolution around the Sun. These are predictable movements. With careful observation, scientists can create models of the motion of celestial objects and predict where these objects will be at a later date. In this Investigation, you will construct a device called a planisphere to help you predict the positions of constellations in the night sky.

Question

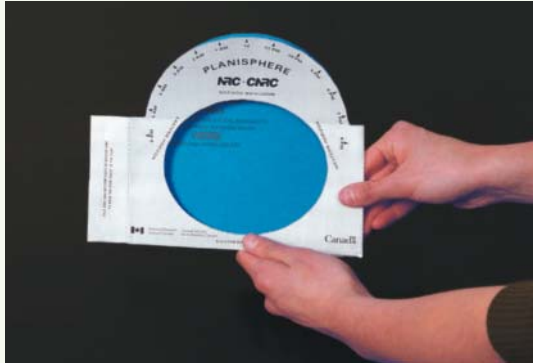
How can you use a planisphere to predict which constellations are visible in the night sky at a certain location and time?

Procedure

1. Obtain the materials you need to make your planisphere.
2. Glue the star map to one side of a file folder and cut it out.
3. Glue the star map frame to the second file folder so that the spine of the file folder is at the bottom of the frame.
4. Cut around the edge of the star map frame through both layers of the file folder.
5. Cut out the inside of the star map frame through only the upper side of the file folder, so the backing is still uncut and shows through the inside of the frame.



6. Wrap the tabs around the back of the frame and glue them in place to the backing.



7. Slide the star map into the frame so that the constellations are visible through the frame.



8. Make a small hole through the star map and the backing at the location of the star Polaris.
9. Insert the split pin through the hole and fasten it so that the map can spin around this point.



Process and Analyze

1. Why do the constellations appear to rotate around Polaris?
2. Rotate the star map so that April 15 lines up with 8 P.M. Which constellations are visible on the southern horizon? Which ones are directly overhead?
3. Rotate the star map so that October 15 lines up with 8 P.M. How has the sky changed from your observations of April 15?
4. Rotate the star map so that today's date lines up with 8 P.M. Slowly rotate the wheel through 9 P.M., 10 P.M., and so on until the morning hours are visible. Describe the change in the position of the constellations throughout the night.

Evaluate and Communicate

5. Predict which constellations will be visible on the southern horizon during the winter solstice and the summer solstice. Use your planisphere to verify your predictions.
6. Rotate the star map so that it matches today's date. Which constellations will be visible tonight in the southern sky at 10 P.M.?
7. Why are the Sun and the planets not included on a planisphere?

Apply and Innovate

8. You can use the Big Dipper to tell time. Use your planisphere and your own ingenuity to help you figure out how this could be done. Check your ideas by researching online or using print reference materials.